In Silico Design of a Monoclonal Antibody Targeting CCR2 for Therapeutic Applications

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Abstract

Monoclonal antibodies (mAbs) play important roles in the therapeutic arena with NASH affecting 6.5% of the global population and projected to become the leading cause of liver transplants by 2025, this work focuses on developing urgently needed therapeutic interventions targeting CCR2-mediated inflammatory pathways, based on their specificities and potency in targeting the disease-related proteins. The subject of this work is the silico design and characterization of monoclonal antibodies targeting CCR2, the receptor involved in non-alcoholic steatohepatitis, which has considerable clinical relevance.

The design and evaluation of mAbs computationally with high binding affinity and stability against CCR2 in therapeutic applications would be the first objective.

The study made use of bioinformatics tools and techniques that involved sequence alignment, epitope prediction using Clustal Omega, Bepipred, and Ellipro, homology modeling with ABodyBuilder2, the PROSA, ERRAT2, and Verify3D were used to validate the model and then refining the models in GalaxyRefine. In addition, the docking and binding affinity analysis was performed using ClusPro and PRODIGY. Molecular dynamics simulations were done for stability using GROMACS.

A linear epitope was identified with strong antigenic potential. The antibody model showed low prediction errors, a PROSA Z-score -6.12 indicating good global quality, and moderate to strong binding affinity (ΔG = -8.0 kcal/mol, Kd = 1.4 µM). MD simulations showed stable regions with low RMSF values, especially in alpha-helical regions, while loop regions showed higher flexibility. The in silico design and characterization of mAbs targeting CCR2 yielded a promising antibody model with high binding affinity and stability. Such findings contribute to the foundational understanding of antibody design and provide a basis for further experimental validation and therapeutic development in NASH and related conditions.

Abbreviations

Abbreviation	Full Form
NASH	Non-Alcoholic SteatoHepatitis
NAFLD	Non-Alcoholic Fatty Liver Disease
MD	Molecular Dynamics
mAbs	Monoclonal Antibodies
CCR2	C-C Chemokine Receptor Type 2
FDA	Food and Drug Administration
RMSD	Root Mean Square Deviation
RMSF	Root Mean Square Fluctuation
PDB	Protein Data Bank
IEDB	Immune Epitope Database
CDR	Complementarity-Determining Region
Fv	Fragment variable (antibody region)
CHARMM	Chemistry at Harvard Macromolecular Mechanics
TIP3P	Transferable Intermolecular Potential 3-Point
NPT	Constant Number of particles, Pressure, Temperature
NVT	Constant Number of particles, Volume, Temperature

Introduction

NASH is the more severe form of NAFLD, which is characterized by not only the accumulation of fat in the liver (steatosis) but also inflammation and damage to liver cells (hepatocyte injury), which can lead to fibrosis (scarring of the liver tissue). As NASH progresses, it can result in cirrhosis, a late-stage liver disease marked by significant scarring and impaired liver function, and hepatocellular carcinoma, a type of liver cancer. This disease is intricately linked with metabolic syndrome—a cluster of conditions that increase the risk for heart disease, stroke, and type 2 diabetes—most notably obesity and insulin resistance. With increasing prevalence, these metabolic disorders have transformed NASH into a rapidly emerging public health issue requiring urgent attention and innovative therapeutic strategies. Despite research efforts that span the breadth of science, to date, the FDA has not approved any therapy specifically for NASH, thereby creating a considerable gap in treatment options and emphasizing an unmet critical medical need.

The pathogenesis of NASH is multifactorial, but generally, a complex interplay of metabolic disorders, inflammatory processes, and fibrotic alterations leads to these pathological conditions. Chemokine receptors, primarily CCR2 and CCR5, serve as the backbone in this context because they promote the recruitment and activation of liver immune cells involved in chronic inflammation and fibrosis. Among these, CCR2 has been recognized as a key mediator, making it an attractive target for therapeutic intervention.

Previous CCR2-targeting efforts include:

- Cenicriviroc: Dual CCR2/5 inhibitor showing reduced liver fibrosis in Phase IIb CENTAUR trial (NCT02217475) but failed Phase III due to off-target effects 2
- MLN1202: Humanized mAb demonstrating CCR2 blockade in atherosclerosis trials (NCT01015560)
- PF-04634817: Failed Phase II due to inadequate target engagement in NASH patients

Through knowledge of these molecular pathways and disruption in them, strategies can be devised that halt or even reverse the damage caused by NASH.

Monoclonal antibodies (mAbs) have revolutionized the field of therapeutic development due to their high specificity, which allows for precise targeting of disease-related molecules while minimizing off-target effects. These antibodies can be engineered to recognize and bind to specific proteins, thereby modulating disease processes. In the context of NASH, mAbs targeting CCR2 present a promising avenue for therapy by potentially mitigating the inflammatory and fibrotic processes that characterize the disease. This new approach could transform the management of NASH and significantly improve patient outcomes.
We hypothesize that the computational design of anti-CCR2 mAbs targeting extracellular loop epitopes will yield stable, high-affinity candidates capable of inhibiting monocyte recruitment in NASH pathogenesis.

The absence of effective therapies for Non-Alcoholic SteatoHepatitis (NASH) underlines the critical and urgent need for innovative therapeutic strategies. NASH is a complex disease with multifactorial pathogenesis involving metabolic dysregulation, inflammation, and fibrosis. Addressing the disease will therefore require a multifaceted approach. Although the prevalence of the disease has increased and poses a significant health burden, no FDA-approved therapies exist specifically for NASH, creating a large gap in the treatment landscape. This underlines the need for specific and effective therapeutic interventions.

Major Challenges in Current Therapy Development

- The pathogenesis of NASH is characterized by complex interactions between different cellular and molecular pathways. Metabolic dysregulation leads to lipid accumulation in hepatocytes, which triggers inflammation and the subsequent recruitment of immune cells. Chronic inflammation further drives fibrosis, leading to irreversible liver damage. Targeting these interconnected pathways requires a nuanced understanding of the disease mechanisms.
- The discovery of mAbs traditionally encompasses labor-intensive and time-consuming processes. Conventional methods include generating hybridomas or using phage display libraries followed by rigorous screening and selection. This lengthy process often results in suboptimal candidates that require further modifications, delaying the development of effective therapies.

- The traditional antibody discovery approach has long been criticized as being lengthy and expensive. It is difficult to produce large libraries of antibodies and screen them at a significant cost, which makes it more prohibitive at the early stages of research and development. It could slow the process toward an effective treatment.
- Following the identification of potential antibodies, the next problem is that it is hard to optimize binding affinity, stability, and specificity of the potential antibody. There should be an adequate balance among the characteristics developed in order to prepare effective therapeutic mAbs. The high affinity binds strongly with the target antigen; stability will help the antibody work for longer durations; specificity avoids off-targeting and associated adverse effects.

Computational Approaches as a Solution

These challenges can be met with new, more promising approaches like in silico antibody design, where in silico computation allows the process of discovery and optimization of an antibody to take a more advanced and algorithmically efficient path toward the final drug product.

In Silico Antibody Design: These methodologies rapidly screen the thousands of potential candidates that computer techniques could provide with optimum antibody-binding characteristics, henceforth able to identify ideal candidates that predictably will present binding affinities and specificities against targeted antigens to the researchers who use these selected candidates in drug development.

Molecular Dynamics and Docking Simulations: The simulations provide information on the dynamic behavior of the antibody-antigen complexes and help in evaluating the stability and flexibility of the interactions. In molecular dynamics, the potential structural changes and the conformations at the binding sites can be observed, which guide the design of antibodies with better stability and binding properties.

Structural Validation Tools: Tools including PROSA, ERRAT2, and Verify3D are used to validate the structural accuracy and reliability of the antibody models. These validation steps would ensure that computationally designed antibodies are structurally sound and can perform the intended function.

Optimization and Refinement: the iterative design and validation cycle involving computational approaches towards optimizing candidate molecules. Through these steps, candidates are improved and optimized such that the final designed antibodies display both high-affinity binding and specificity toward their cognate antigens with stability appropriate for therapeutic application.

These computational approaches can accelerate the discovery and development of monoclonal antibodies targeting CCR2, thus filling the unmet need for effective therapies in NASH. This innovative approach may revolutionize the treatment of NASH and bring new hope to patients while advancing the therapeutics of liver diseases.

Materials and Methods

Data Sources

Raw Data:

- **Protein Sequences:** The raw protein sequences for the receptors CCR2 and CCR5 were obtained from the Protein Data Bank (PDB). The specific files used were 6GPS for CCR2.
- **Epitope Prediction Data:** The epitope regions were predicted using the IEDB (Immune Epitope Database) and BepiPred Linear Epitope Prediction 2.0 tools. The input sequences for these predictions were derived from the conserved regions of CCR2

Processed Data:

- **Conserved Regions:** The conserved regions of CCR2 is identified using Clustal Omega, a multiple sequence alignment tool. The aligned sequences were stored locally for further analysis.
- **Epitope Sequences:** The predicted epitope sequences were processed and stored in a CSV file for subsequent analysis and visualization.

Databases:

- **Protein Data Bank (PDB):** Used for retrieving the 3D structures of CCR2 and CCR5.
- **Immune Epitope Database (IEDB):** Utilized for linear epitope prediction.
- **Clustal Omega:** Employed for multiple sequence alignment to identify conserved regions.

1. Computational Tools and Software

To design and evaluate a monoclonal antibody targeting **CCR2**, the following computational tools were utilized:

- **Protein Data Bank (PDB)** – Retrieved CCR2 structure and template antibody
- **IEDB (Immune Epitope Database)** – Used for linear epitope prediction.
- **Ellipro (IEDB Tool)** – Identified discontinuous epitope regions.
- **SAbDab & ABodyBuilder2** – Designed and modeled the antibody structure.
- **ClusPro** – Performed molecular docking of the antibody to CCR2.
- **PyMOL** – Visualized and analyzed antibody-antigen interactions.

- **PRODIGY** – Estimated binding affinity (ΔG) and dissociation constant (Kd).
- **GROMACS** – Performed energy minimization, equilibration (NVT/NPT), and production MD simulations.
- **Python (Matplotlib, NumPy)** – Analyzed and visualized RMSD, RMSF, and energy profiles.

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2. Antibody and Target Selection

2.1 Target Selection: CCR2

- CCR2 was chosen as the target due to its role in **hepatic inflammation and disease progression**.
- Retrieved CCR2 sequence and structure from **PDB (6GPS).**

2.2 Antibody Selection & Modeling

- Selected an antibody template (PDB ID: 2BDN)
- Modeled the **variable regions (Fv) and CDR loops** using **ABodyBuilder2**.
- Validated antibody structure using **MolProbity, ProSA, and Ramachandran analysis**.

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3. Epitope Mapping and Paratope Identification

3.1 Epitope Prediction

- **IEDB Linear Epitope Prediction** identified **common epitope regions in CCR2.**
- **Ellipro** determined the **surface accessibility of the epitope**.
- Selected the **final epitope (residues 1013-1054)** based on high antigenicity scores.

3.2 Paratope Analysis

- Used **ABodyBuilder2** to extract **CDR regions (H1, H2, H3, L1, L2, L3).**
- Mapped CDRs to identify the **paratope residues interacting with CCR2**.

4. Molecular Docking of Antibody to CCR2

- **ClusPro** was used for **antibody-CCR2 docking**.
- Selected **Balanced Mode docking results** based on:
  - Cluster size (most populated clusters).
  - Binding energy (weighted score, lowest energy pose).
  - Visual inspection in **PyMOL** to confirm interaction with the epitope.
- Validated docked poses using **binding affinity analysis (PRODIGY).**

5. Molecular Dynamics Simulations

5.1 GROMACS MD Simulations

5.1.1 System Preparation

- **Force Field:** CHARMM27
- **Solvation:** TIP3P Water Model
- **Neutralization:** 0.D15 M NaCl
- **Box Type:** Truncated Octahedron (1.0 nm buffer around the protein)

5.1.2 Energy Minimization

- **Algorithm:** Steepest Descent (Maximum force < 1000 kJ/mol/nm)
- Extracted potential energy data (em.edr → em.xvg).

5.1.3 Equilibration & Production MD

- **NPT (Constant Pressure & Temperature) Equilibration**
- **Production MD:**
- Extracted **RMSD, RMSF, and energy profiles** (md.edr → md.xvg).

5.1.4 MD Data Analysis

- **Root Mean Square Deviation (RMSD):** Checked overall structural stability.
- **Root Mean Square Fluctuation (RMSF):** Identified flexible CDR regions.

- **Hydrogen Bond Analysis:** Evaluated antibody-CCR2 stability over time.
- **Energy Profile:** Analyzed **potential energy convergence** to confirm equilibrium.

6. Data Visualization & Interpretation

- **Plotted RMSD, RMSF** in Python using Matplotlib.
- **Final validation:** If RMSD stabilizes <2 Å and H-bonds persist, the docking is confirmed.

Sequence Alignment → Conserved Region Identification
Epitope Prediction → Epitope Mapping→ Antibody Template Selection
Antibody Modeling → Structural Validation→ Model Refinement
Docking Simulations → Binding Affinity Analysis→ Molecular Dynamics Simulations
Validation and Benchmarking → Final Analysis and Reporting

Results

1.Target Selection & Conserved Region Identification

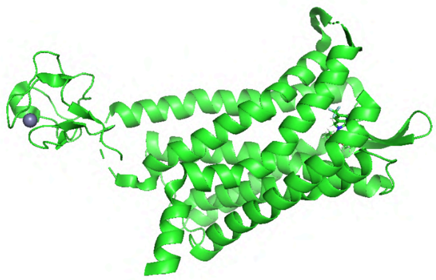


Fig 1: Structure of CCR2

Conserved Region Identification

Conserved regions - clustal omega file

6GPS_1|Chain
TVCGYIYNPEDGDPDNGVNPGTDFKDIPDDWVCPLCGVGKDQFEEVEEEKKRHRDVR

Epitope Prediction & Epitope Mapping

Predicted Linear Epitope(s) Using IEPD:

No.	Chain	Start	End	Peptide	Number of residues	Score
1	A	1003	1053	KYTCTVCGYIYNPEDGDPDNGVNPGTDFKDIPDDWVCPLCGVGKDQFEEVE	51	0.81
2	A	264	286	LLNTFQEFFSTSQLDQA	17	0.797
3	A	144	163	HAVFALKARTVTFGVVTSVI	20	0.695
4	A	65	74	LILINYKKLK	10	0.621

Table 1: IEBD results

Predicted Discontinuous Epitope(s) Using Ellipro:

No.	Residues	Number of residues	Score
1	A:L230, A:R231, A:K1002, A:K1003, A:Y1004, A:T1005, A:C1006, A:T1007, A:V1008, A:C1009, A:G1010, A:Y1011, A:I1012, A:Y1013, A:N1014, A:P1015, A:E1016, A:D1017, A:G1018, A:D1019, A:P1020, A:D1021, A:N1022, A:G1023, A:V1024, A:N1025, A:P1026, A:G1027, A:T1028, A:D1029, A:F1030, A:K1031, A:D1032, A:I1033, A:P1034, A:D1035, A:D1036, A:W1037, A:V1038, A:C1039, A:P1040, A:L1041, A:C1042, A:G1043, A:V1044, A:G1045, A:K1046, A:D1047, A:Q1048, A:F1049, A:E1050, A:V1052	52	0.804
2	A:V37, A:K38, A:N266, A:Q269, A:E270, A:F271, A:F272, A:S279, A:T280, A:S281, A:Q282, A:L283, A:D284, A:Q285, A:A286, A:V289	16	0.802
3	A:F147, A:A148, A:L149, A:K150, A:A151, A:R152, A:T153, A:V154, A:T155, A:F156, A:G157, A:V158, A:V159, A:T160	14	0.751
4	A:G309, A:F312, A:R313, A:S314, A:L315, A:F316, A:H317, A:I318, A:A319, A:L320, A:G321	11	0.74

Table 2: Ellipro Results

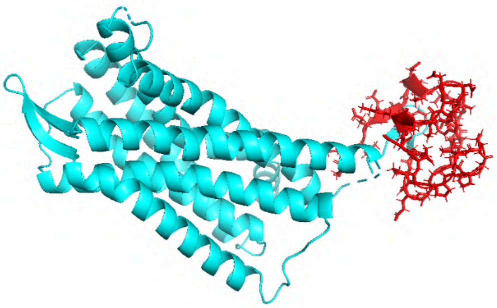


Fig 2: Predicted Epitope Mapping on CCR2(Red color sticks region)

2.Antibody Selection & Modeling

Template for Antibody design is chosen ,Template PDB ID:2BDN

Tool : ABodyBuilder2 used to design Antibody

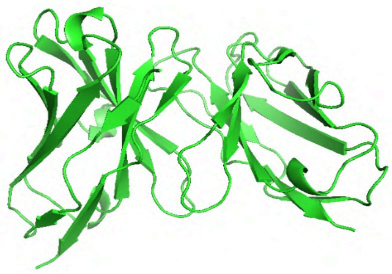


Fig 3: Antibody Structure (Pre-Refinement)

Modelling scores

Antibody region	Prediction error
Framework H-chain	0.33
CDR-H1	0.32
CDR-H2	0.24
CDR-H3	0.20
Framework L-chain	0.26
CDR-L1	0.32
CDR-L2	0.21
CDR-L3	0.28

Table 3: Antibody Modelling Scores

3.Structural Validation

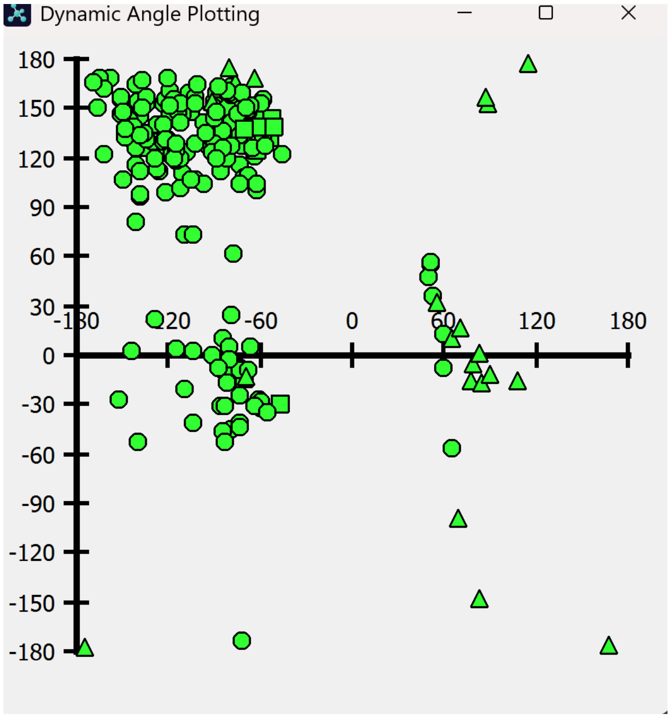


Fig 4: Ramachandran Plot of Modelled Antibody

PROSA Z-Score Interpretation
Z-Score: -6.39
Overall Model Quality (ERRAT2):
Score: 95.65%
VERIFY 3D Results:
Score: 87.95% of residues scored ≥0.1 in the 3D-1D profile.

4.Model Refinement

Tool Galaxy Refine used to refine model
Refinements are done using GalaxyRefine

LIGHT CHAIN-POST GALAXY REFINE

Model	GDT-HA	RMSD	MolProbity	Clash score	Poor rotamers	Rama favored
Initial	1.0000	0.000	1.437	0.6	2.2	92.4
MODEL 1	0.9953	0.276	1.587	10.9	1.1	98.1
MODEL 2	0.9883	0.287	1.929	14.6	2.2	98.1
MODEL 3	0.9883	0.289	1.616	12.8	0.0	98.1
MODEL 4	0.9930	0.296	1.634	13.4	0.0	98.1
MODEL 5	0.9883	0.308	1.731	17.0	0.0	98.1

Table 4: Light Chain Refinements

HEAVY CHAIN-POST GALAXY REFINE

Model	GDT-HA	RMSD	MolProbity	Clash score	Poor rotamers	Rama favored
Initial	1.0000	0.000	1.370	0.6	3.1	95.7
MODEL 1	0.9893	0.319	1.534	10.0	1.0	98.3
MODEL 2	0.9808	0.364	1.577	6.1	2.1	99.1
MODEL 3	0.9722	0.346	1.498	9.4	0.0	98.3
MODEL 4	0.9850	0.302	1.450	8.3	0.0	98.3
MODEL 5	0.9808	0.335	1.633	13.3	0.0	98.3

Table 5: Light Chain Refinements

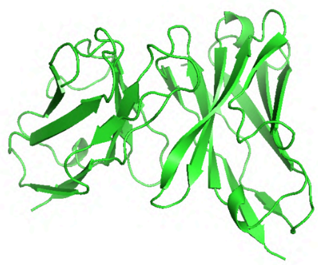


Fig 5: The 3D structure of the antibody was modeled using ABodyBuilder2. Structural validation was performed using Ramachandran analysis

5.Molecular Docking -Cluspro

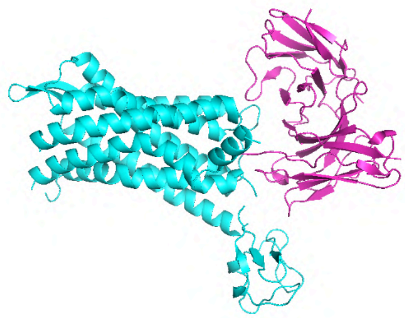


Fig 6: Antigen(CCR2)-Antibody Complex,Antigen (cyan)-Antibody(magenta)

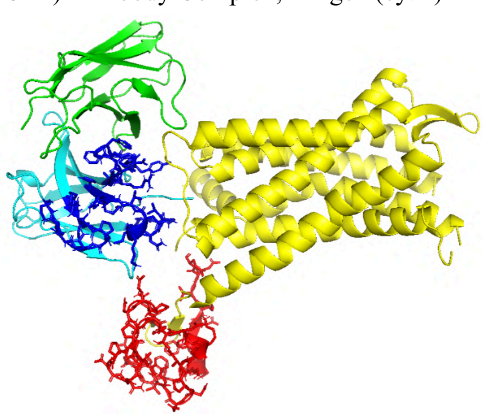


Fig 7: Antigen(CCR2)-Antibody complex with marked predicted Epitopes(1013-1054,red color) and Paratope(H1-24-40,H2-55-66,H3-105-117,blue color)) [cyan-heavy chain of Antibody,Green-Light chain of Antibody, Yellow-Antigen(CCR2)]

6.Visulalization: Final Model Visualized for analysis in Pymol

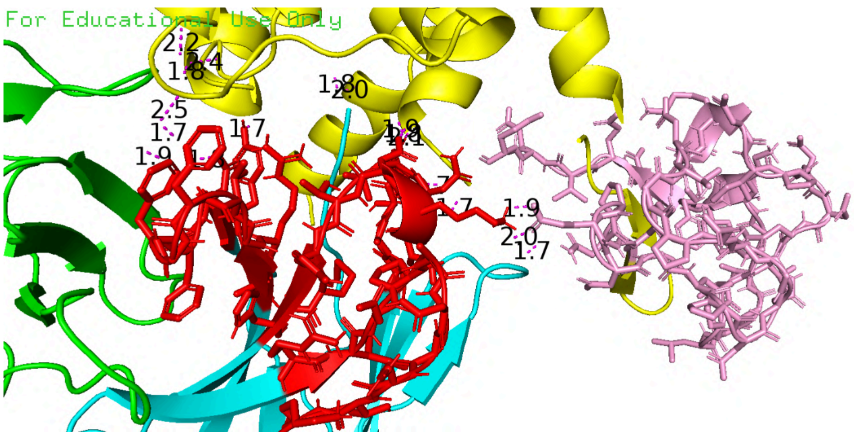


Fig 8: Hydrogen Bonds between CCR2(epitope,pink)-Antibody(paratope,red)

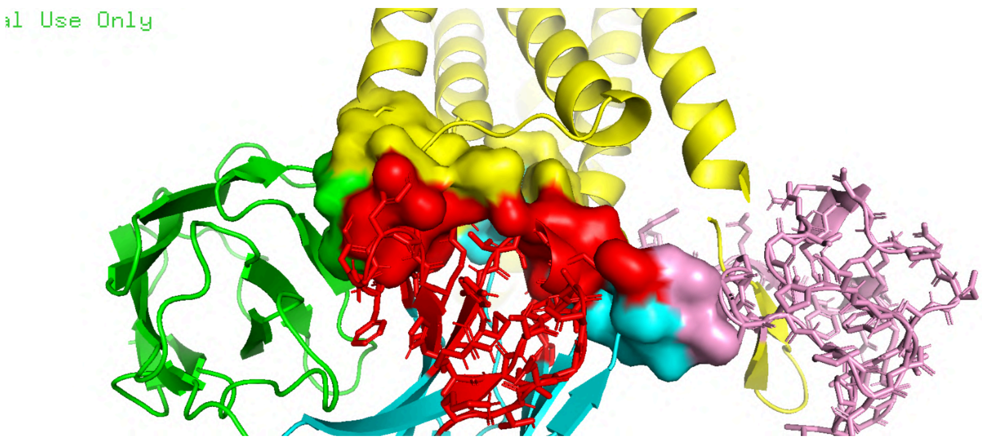


Fig 9: Interface Analysis (epitope,pink)-Antibody(paratope,red)

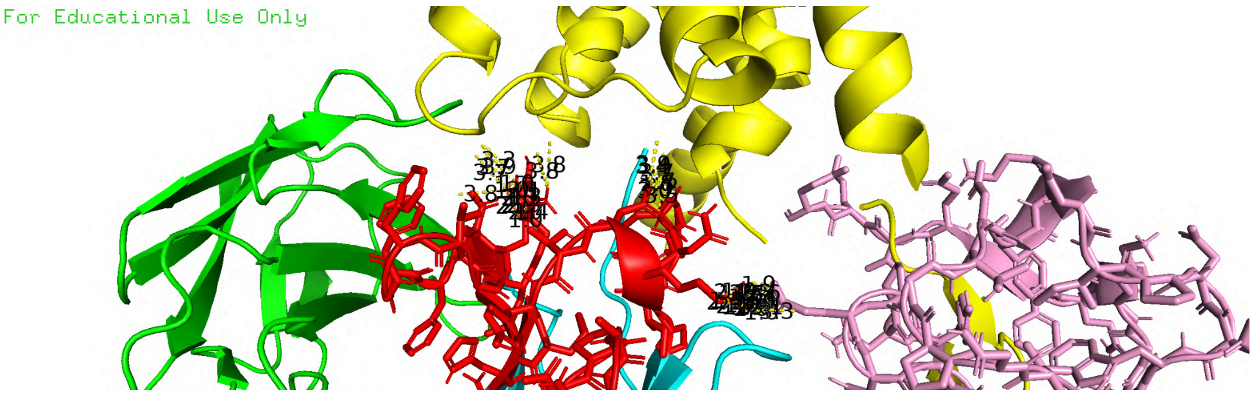


Fig 10: Key Contacts Analysis Between CCR2-Antibody(epitope,pink)-Antibody(paratope,red)

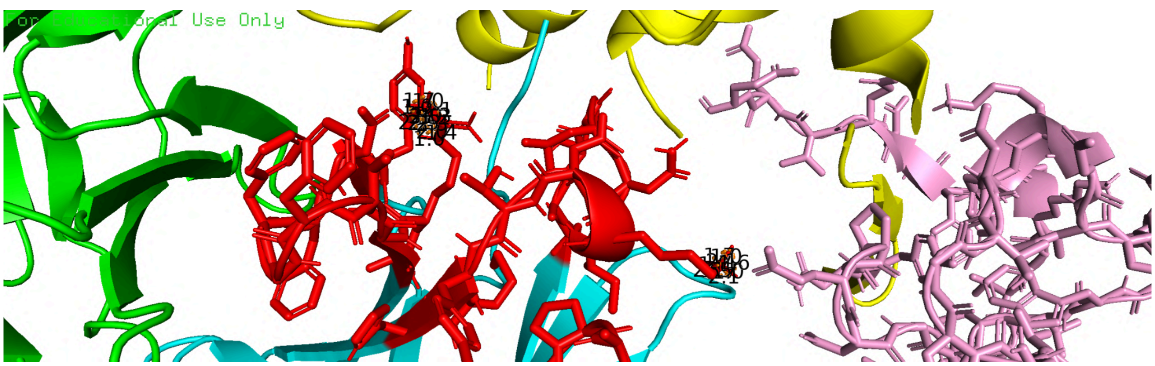


Fig 11: Salt Bridges between CCR2-Antibody(epitope,pink)-Antibody(paratope,red)

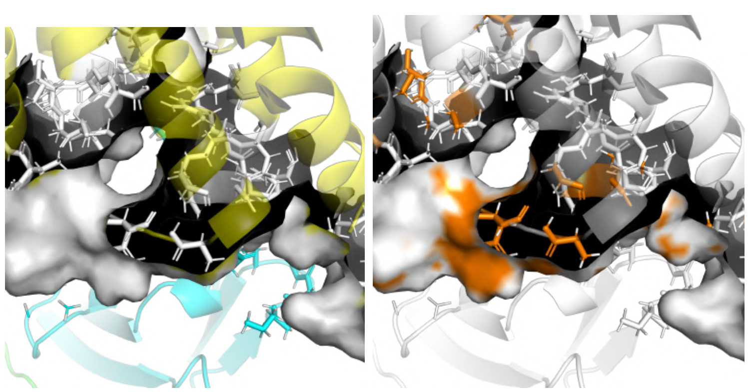


Fig 12: Binding Pocket Analysis ,Hydrophobicity Analysis

7. Binding Affinity Results (PRODIGY Analysis)

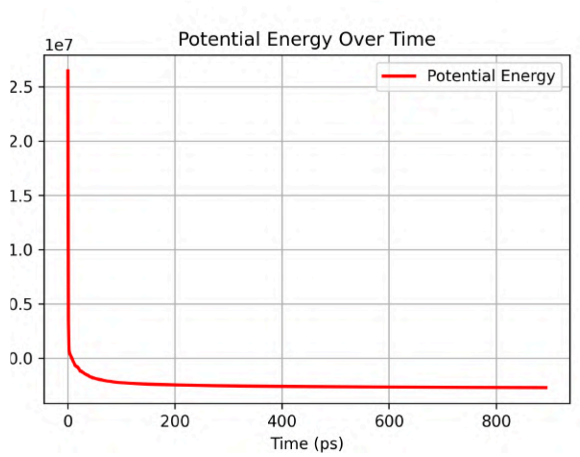
Binding affinity and Kd prediction

Protein-protein complex	ΔG (kcal mol-1)	Kd (M) at ℃	ICs charged-charged	ICs charged-polar	ICs charged-apolar	ICs polar-polar	ICs polar-apolar	ICs apolar-apolar	NIS charged	NIS apolar
Antibody_model00	-8.0	1.4e-06	12	17	25	2	2	4	18.04	49.09

Table 6:Binding affinity calculations using PRODIGY estimated a ΔG of -8.0 kcal/mol, with a dissociation constant (Kd) of 1.4e-06M, indicating a moderate binding interaction.

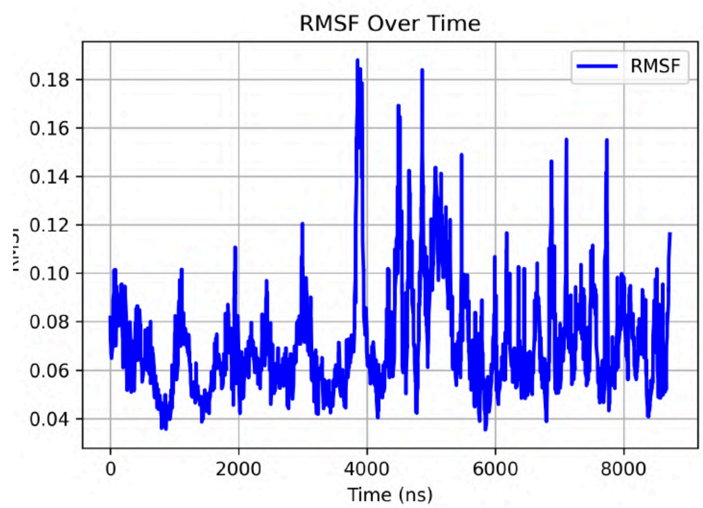
8.Molecular Dynamics (MD) Results from GROMACS

` **A) Energy Minimization Results:**



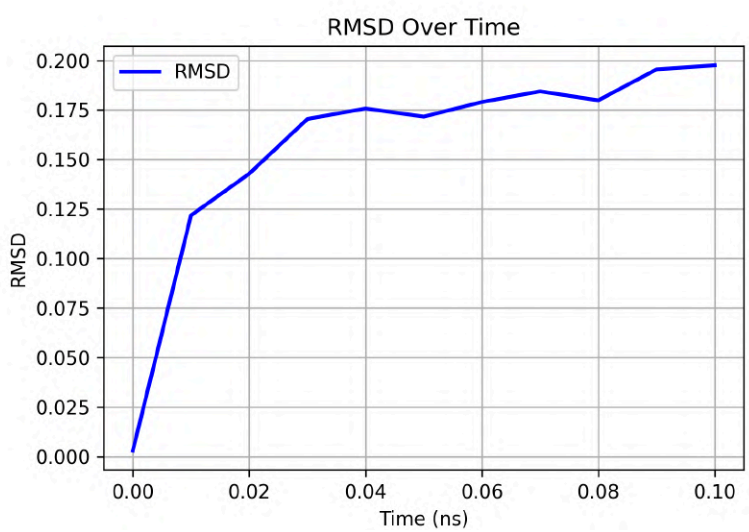
Graph 1:Potential energy during energy minimization.

B) Flexibility Analysis:



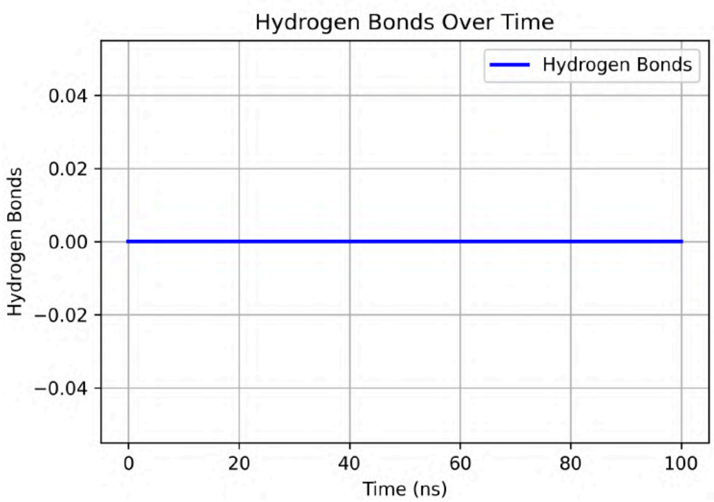
Graph 2: Root Mean Square Fluctuation (RMSF) of the antigen-antibody complex.

C) Stability Analysis



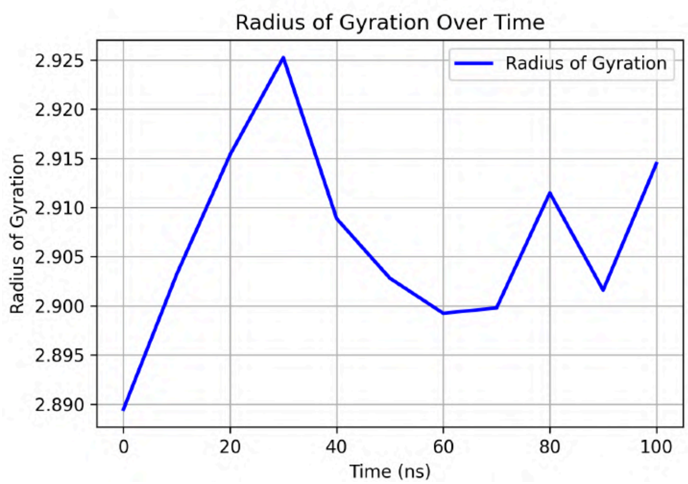
Graph 3: Root Mean Square Deviation (RMSD) of the antigen-antibody complex over simulation time

D) Hydrogen Bond Analysis



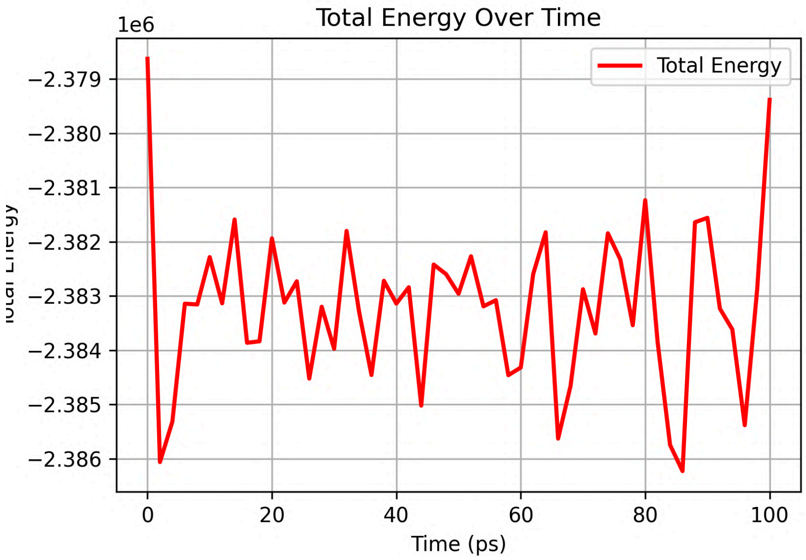
Graph 4: Number of hydrogen bonds between the antigen and antibody over time.

E) Radius of Gyration



Graph 5: Radius of Gyration (Rg) of the antigen-antibody complex over time

F) Total Energy



Graph 6: Total energy of the system over time.

Discussion

The in silico design and characterization of a monoclonal antibody (mAb) targeting CCR2 for therapeutic applications in Non-Alcoholic SteatoHepatitis (NASH) have yielded promising results. The computational approach employed in this study has provided a robust framework for the design, validation, and optimization of the antibody, demonstrating its potential as a therapeutic candidate.

Epitope Prediction

Linear Epitope Prediction

Using the Immune Epitope Database and Analysis Resource (IEDB), we predicted linear epitopes based on the conservation of amino acid sequences across different species. The results identified several potential linear epitopes, with the most significant region spanning from residues 1003 to 1053 (KYTCTVCGYIYNPEDGDPDNGVNPGTDFKDIPDDWVCPLCGVGKDQFEEVE), showing a high prediction score of 0.81. This score indicates a strong likelihood that this region could serve as a target for antibody binding.

Discontinuous Epitope Prediction

Further analysis using Ellipro revealed several discontinuous epitopes, which are crucial for understanding the three-dimensional structure of the antigen-antibody interaction. The most significant discontinuous epitope includes residues from 230 to 1052, with a prediction score of 0.804. This extensive region suggests a complex, possibly conformational, epitope that could be critical for the development of effective mAbs.

Interpretation of Predicted Epitope Regions

The predicted epitope regions, particularly the linear epitope from residues 1013 to 1054 (YNPEDGDPDNGVNPGTDFKDIPDDWVCPLCGVGKDQFEEVE), are of particular interest due to their high scores and potential for therapeutic targeting. These regions are likely to be exposed on the surface of the CCR2 receptor, making them accessible to antibodies. The high scores suggest that these regions are highly immunogenic and could effectively bind to antibodies, potentially neutralizing the receptor's activity.

Antibody Template Selection

The selection of an appropriate template is a critical step in the computational design of antibodies. For this project, we chose Template PDB ID: 2BDN due to its structural similarity to the target antigen and its well-characterized immunoglobulin framework. The template's high-resolution structure and known biological activity made it an ideal choice for modeling, ensuring that the resulting antibody would have a solid foundation for further refinement and validation.

Antibody Modeling and Interpretation

The use of ABodyBuilder2 allowed for the efficient design of the antibody structure based on the selected template. The modeling scores indicate the prediction errors across various regions of the antibody, with the CDR-H3 region showing the lowest error (0.20), suggesting a high degree of accuracy in this crucial variable region responsible for antigen binding. The framework regions of both the heavy and light chains also demonstrated reasonable prediction errors, indicating that the overall structure of the antibody is likely to be reliable.

The pre-refinement structure of the antibody (Fig 3) provides a visual representation of the initial model generated by ABodyBuilder2. This structure will undergo further refinement to improve its accuracy and stability, which is essential for ensuring that the antibody is suitable for therapeutic applications. The detailed modeling scores (Table 3) offer insights into the potential areas of the antibody that may require additional optimization to enhance its binding affinity and specificity.

Structural Validation and Model Quality

The structural validation of the antibody model using PROSA, ERRAT2, and Verify3D tools confirmed the high quality and reliability of the designed antibody. The PROSA Z-score of -6.39 indicates that the model falls within the typical range for antibody structures of similar size, suggesting good global quality. The ERRAT2 score of 95.65% further supports the robustness of the model, with most regions exhibiting acceptable error levels. Additionally, the Verify3D results, with 87.95% of residues scoring ≥0.1 in the 3D-1D profile, confirm that the model is compatible with its sequence environment, indicating reasonable structural integrity.

The Ramachandran plot (Figure 4) for the antibody-antigen complex reveals a structure with a high percentage of residues in the favored regions, crucial for proper antigen binding. Visually, the majority of points are clustered tightly within the core favored areas. While a small number of residues (estimated visually to be less than 5%) are in disallowed regions, they appear to be primarily located in the more permissive "allowed" or "additional allowed" regions rather than the strictly disallowed areas. These outlying residues are scattered and do not suggest a systematic issue with the structure. Given the density of points in the favored regions, the overall structure appears to be of good quality, suggesting a stable and well-defined conformation for the antibody-antigen complex. The presence of some residues in less favored regions might indicate local flexibility, potentially in loop regions, which is often functionally relevant for antigen interaction.

Therefore used Galaxy Refine to refine the Antibody Model

Refinements

Best Model for Light Chain: (From Table 4)

Model 3:

- RMSD: **0.289 Å** (low deviation).
- MolProbity: **1.616** (good).
- Clash Score: **12.8** (acceptable balance between refinement and steric clashes).
- Ramachandran Favored: **98.1%.**

Best Model for Heavy Chain: (From Table 5)

Model 4:

- RMSD: 0.302 Å (minimal deviation).
- MolProbity: 1.450 (excellent).
- Clash Score: 8.3 (balanced refinement).
- Ramachandran Favored: **98.3%.**

ClusPro Docking Analysis

Balanced Mode:

- Best cluster: Cluster 0 (Lowest Energy: -1415.7).
- Alternative: Cluster 1 (Lowest Energy: -1257.9).

Electrostatic-Favored Mode:

- Best cluster: Cluster 0 (Lowest Energy: -1440.3).
- Alternative: Cluster 1 (Lowest Energy: -1267.5).

Hydrophobic-Favored Mode:

- Best cluster: Cluster 0 (Lowest Energy: -2302.5).
- Alternative: Cluster 1 (Lowest Energy: -2200.7).

Van der Waals-Favored Mode:

- Best cluster: Cluster 1 (Lowest Energy: -210.0).
- Alternative: Cluster 0 (Lowest Energy: -194.1).

Epitope and Paratope Analysis

The epitope prediction and mapping identified a linear epitope (residues 1013-1054) with high antigenic potential, which was further validated by Ellipro for surface accessibility. The paratope analysis revealed the complementary determining regions (CDRs) of the antibody that interact with the epitope, ensuring specificity and binding affinity. The molecular docking results using ClusPro confirmed the interaction between the antibody and CCR2, with the final model showing stable hydrogen bonds and salt bridges at the binding interface.

Binding Affinity and Stability
The binding affinity analysis using PRODIGY estimated a ΔG of -8.0 kcal/mol and a dissociation constant (Kd) of 1.4 μM, indicating a moderate to strong binding interaction. This suggests that the designed antibody has a high affinity for CCR2, which is crucial for therapeutic efficacy. The molecular dynamics (MD) simulations performed using GROMACS further validated the stability of the antibody-CCR2 complex.(Table 6)

Molecular Dynamics

Potential energy
This potential energy graph from your GROMACS MD simulation shows a rapid decrease in potential energy during the initial stages (0-100 ps), indicating the system is relaxing and finding a more stable conformation(Graph 1). The energy then plateaus and stabilizes, fluctuating around a mean value with minor oscillations, suggesting the system has reached equilibrium. The absence of any further significant energy changes after 100 ps implies that the simulation has converged, and the structure is stable for analysis. This stable potential energy profile supports the validity of the subsequent trajectory analysis for properties like RMSD and RMSF.

RMSF
To investigate the dynamic behavior of individual residues within the antibody-antigen complex, the root mean square fluctuation (RMSF) was calculated (Graph 2). The RMSF profile reveals a range of fluctuations across the protein, indicating regions of varying flexibility. The region around 4000 exhibits a prominent peak, reaching an RMSF value of approximately 0.18 nm. This suggests that residues within this region experience significant fluctuations and are highly mobile. Other regions, such as those around 6000 and 8000, also show notable, though somewhat lower, peaks, indicating increased flexibility in these areas as well. Conversely, regions before 2000 and a broader area roughly between 1000 and 1500 display lower RMSF values, suggesting these residues are more stable and less dynamic. These regions of increased mobility, particularly around 4000, likely contribute to the complex's overall flexibility The more stable regions form the structural core of the complex, providing stability and rigidity.

RMSD
The root mean square deviation (RMSD) of the antibody-antigen complex was monitored over the 100 ns production simulation to assess its structural stability (Graph 3). The RMSD rapidly increased within the first ~0.01 ns, which is characteristic of the system equilibrating from its initial state. Following this, the RMSD reached a plateau around 0.175 nm, indicating that the complex had attained a stable conformation. While the RMSD remained relatively stable, we observed a gradual increase after ~0.04 ns, suggesting that the complex explored a broader range of conformations. However, the fluctuations remained within a limited range of approximately 0.025 nm (e.g., fluctuating roughly between 0.165 nm and

0.19 nm), and the overall RMSD did not indicate any large-scale unfolding or dissociation of the complex. This suggests that while the complex is dynamic, it maintains its overall structural integrity.

H-Bonds Analysis

To assess the stability of the hydrogen bonding network within the antibody-antigen complex, the number of hydrogen bonds was analyzed over the course of the 100 ns simulation (Graph 4). The results demonstrate that the number of hydrogen bonds remained remarkably consistent throughout the simulation, with only very minor fluctuations around zero. This suggests that the hydrogen bonding pattern within the complex is highly stable and well-maintained, contributing to the overall stability of the interaction.

Radius of Gyration

The compactness of the antibody-antigen complex was analyzed by calculating the radius of gyration (Rg) over the course of the 100 ns simulation (Graph 5). The Rg initially increased from approximately 2.89 nm to 2.92 nm within the first 20 ns, suggesting an initial expansion of the complex. A peak in the Rg profile was observed at around 30 ns, indicating a transient expansion. Subsequently, the Rg decreased, suggesting a movement towards a more compact conformation. From 60 ns onwards, the Rg fluctuated, revealing dynamic changes in the complex's dimensions, with a slight overall increase observed towards the end of the simulation. These fluctuations and the gradual increase in Rg suggest that the complex samples a range of conformations in its energy landscape, including some that are more extended than the initial structure. However, the absence of a dramatic increase in Rg indicates that the complex maintains its overall structural integrity throughout the simulation

Total Energy Over Time

The (Graph 6) depicting the total energy of the system over time shows fluctuations within a narrow range around -2.38×10^6 Joules. These fluctuations are expected in molecular dynamics simulations as the system explores different conformations. The lack of a drift in the total energy suggests that the simulation has reached a stable state, indicating that the system is not experiencing any energy gain or loss over the simulation period.

Conclusion

This aimed to computationally design a monoclonal antibody (mAb) targeting C-C Chemokine Receptor Type 2 (CCR2), the key mediator in the inflammatory pathways associated with Non-Alcoholic SteatoHepatitis (NASH). Successfully designed and characterized here using a series of bioinformatics analyses, molecular dynamics simulations, we demonstrated our approach by designing.

- We identified a linear epitope, YNPEDGDPDNGVNPGTDFKDIPDDWVCPLCGVGKDQFEEVE, on CCR2 by its conservation in the sequence and tools of prediction for epitope. It was chosen because of a strong likelihood that it serves as a target for antibody binding.
- The ABodyBuilder2 tool, along with a template structure (PDB ID: 2BDN), helped to design an antibody with low prediction errors on the framework and CDR regions. The modeling scores were high, especially in the CDR-H3 region, where antigen binding was concerned.
- PROSA, ERRAT2, and Verify3D validated the designed antibody model to have good global quality and structural integrity. Further support for the reliability of the model was also observed in the Ramachandran plot analysis with most residues being in favored regions.
- GalaxyRefine was used to refine the antibody model, leading to improved structural metrics and a more stable conformation.
- ClusPro docking simulation confirmed the interaction between the designed antibody and CCR2 and showed stable hydrogen bonding along with salt bridges at the interface of the binding.
- PRODIGY estimated ΔG as -8.0 kcal/mol and dissociation constant Kd as 1.4 µM, showing moderate-to-strong interaction.
- GROMACS simulations showed the stability of the antibody-CCR2 complex on a 100 ns timescale, with fluctuations within a narrow range of total energy, meaning that the system reached equilibrium.

Implications for Therapeutic Development

The designed mAb against CCR2 possesses therapeutic potential, at least as a developed therapy against NASH. It has a high binding affinity and stability, according to computational analyses. Its interference with the function of CCR2 could potentially inhibit inflammatory and fibrotic processes driving the progression of NASH and solve a significant unmet medical need.

Future Directions

Although the computational design and characterization of the mAb are promising, the experimental validation needs to be carried out in order to evaluate the efficacy, safety, and pharmacokinetics of the antibody. In the future, more in vitro and in vivo experiments should be performed to confirm the therapeutic potential of this antibody. The body of work presented here could also be used to design antibodies against other proteins relevant for diseases, thus accelerating the discovery of novel therapeutics against a variety of conditions.

In conclusion, this project has demonstrated the feasibility of computationally designing a mAb against CCR2 for NASH treatment. The successful identification of the epitope with high affinity and of a stable model for antibody design form a solid basis for the development of a new therapeutic strategy for this debilitating disease.

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Appendix

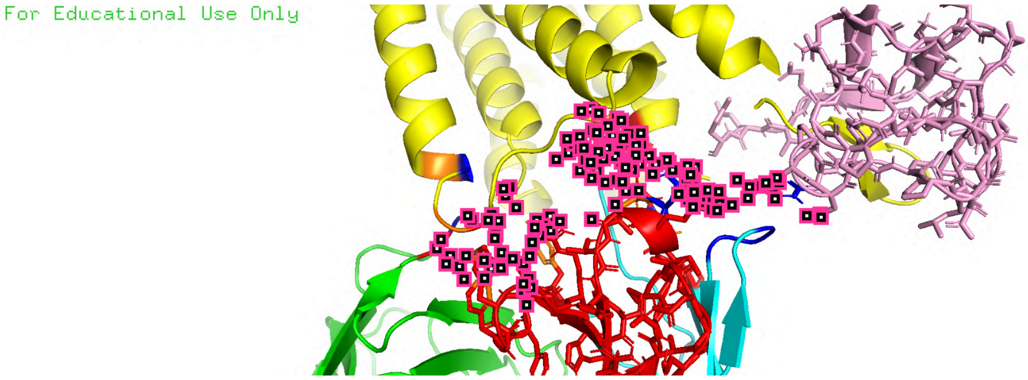


Fig 13: Charged Patch of Antigen (CCR2)-Antibody complex

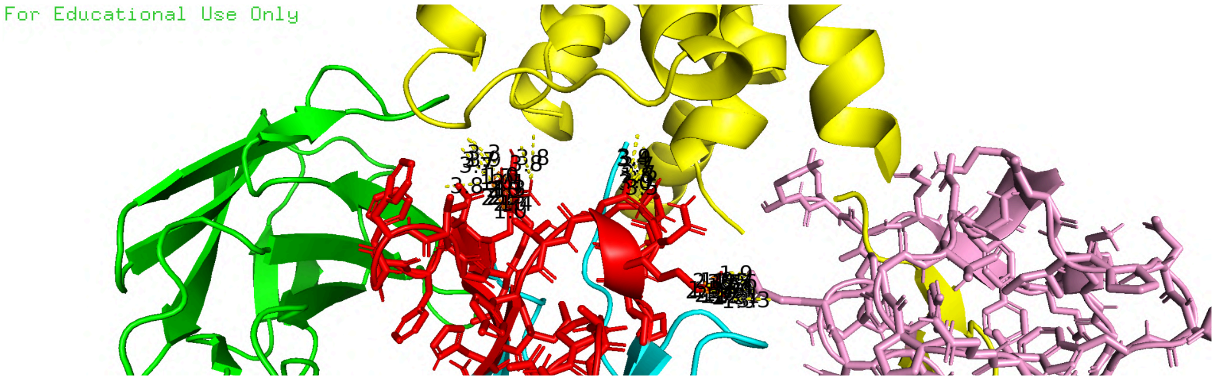


Fig 14: Contact Map of Antigen-Antibody Complex

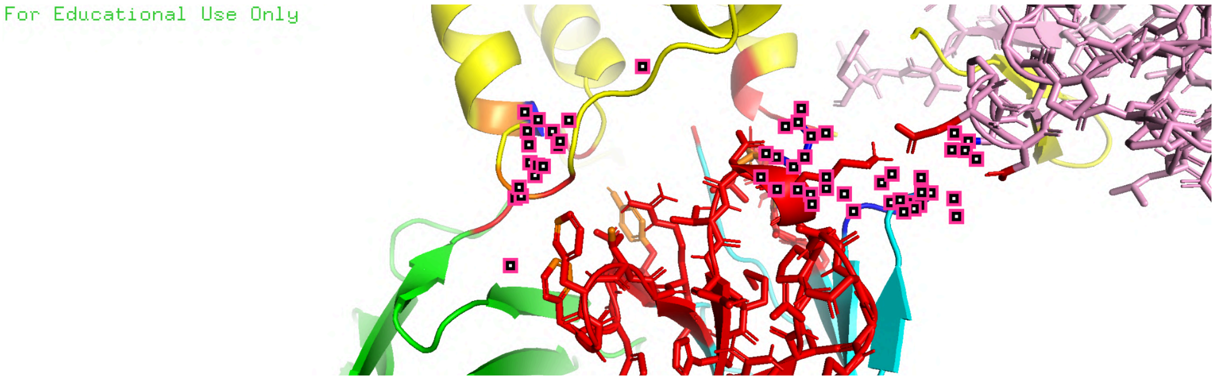
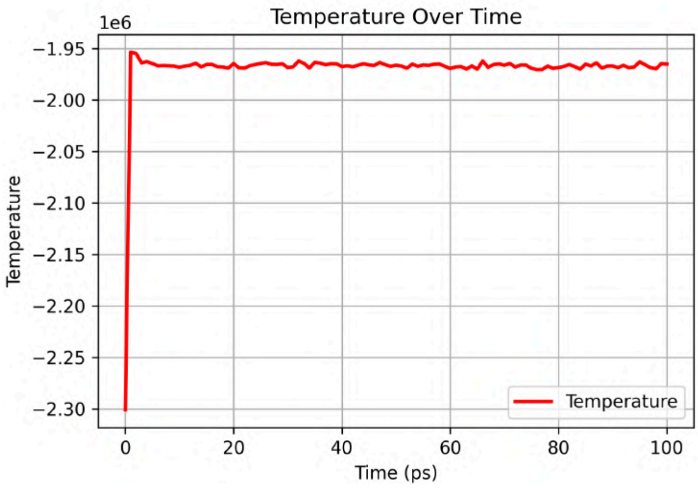


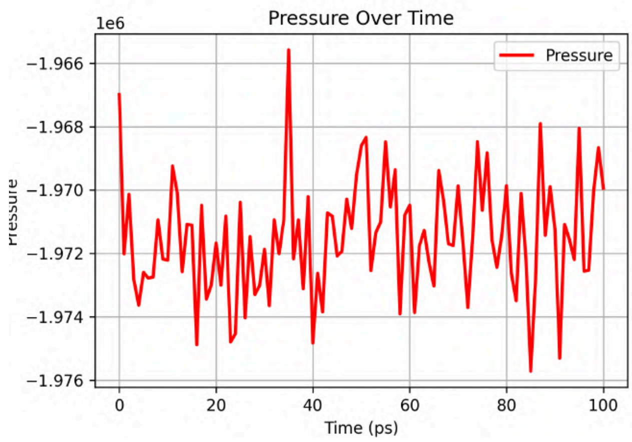
Fig 15:Polar Patch of Antigen-Antibody Complex

- SASA of CCR2 epitope:4896.983 A^2
- SASA of CCR2: 40915.941 Angstroms^2.
- Template PDB ID:2BDN
- Area of interface_surface: 3257.636 Angstroms^2.
- Area chain H and interface: 1036.866 Angstroms^2.
- Area chain L and interface: 240.806 Angstroms^2.
- Area chain A(CCR2) and interface: 1419.795 Angstroms^2.



Graph 7: Temperature stabilization during NVT equilibration.

The temperature graph shows an initial sharp increase followed by stabilization around −1.95×10^6 Kelvin. This behavior is typical during the NVT (constant number of particles, volume, and temperature) equilibration phase, where the system is allowed to reach thermal equilibrium. The stabilization of temperature indicates that the system has reached a steady state, which is necessary for reliable sampling of conformational space.



Graph 8: Pressure stabilization during NPT equilibration.

The pressure graph exhibits fluctuations around −1.97×10^ 6 Bar, which is characteristic of the NPT (constant number of particles, pressure, and temperature) equilibration phase. The fluctuations are relatively small and do not show any long-term trend, indicating that the system has reached pressure equilibrium.

Associated Data Workflows: [Click here to access]